Chapter 5

Food Microbiology

Learning Objectives

After studying this chapter the students will be able to,

- *Know the sources of microorganisms in food*
- *Understand the factors that influence growth of microorganisms in food*
- *Learn about the food spoilage*
- *Appreciate the food preservation methods*
- *Learn and compare the food poisoning and food intoxication*
- *Classify the food borne diseases*
- *Understand the microbial standards and grading of milk*
- *Know the fermented milk products like cheese, yoghurt and curd*

Chapter Outline

5.1 Food Microbiology
5.2 Food Spoilage
5.3 Food Borne Disease
5.4 Food Preservation Methods
5.5 Dairy Microbiology
5.6 Cheese
5.7 Yogurt
5.8 Curd

5.1 Food Microbiology



The field of food microbiology is very broad, encompassing the study of microorganisms which have both beneficial and deleterious effects on the quality and safety of raw and processed foods. The primary tool of microbiologists is the ability to identify and quantitate food-borne microorganisms. Microorganisms in food include bacteria, molds, yeasts, algae, viruses, parasitic worms and protozoans.

Microorganisms are associated with the food we eat in a variety of ways. They may influence the quality of our food. Naturally occurring foods such as fruits and vegetables normally contain some microorganisms and may be contaminated with additional organisms during handling and processing. Food can serve as a medium for the growth of microorganism, and microbial growth may cause the food to undergo decomposition and spoilage. Food may also carry pathogenic microorganisms which when ingested can cause disease. When food with microorganisms that produce toxic substances is ingested, it results in food poisoning. Apart from the pathogenic microorganisms, some microorganisms are used in the preparation and preservation of food products.

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